

## SUMMERLAND, BC, Canada

WMO: 717680

Lat: 49.5625N

Lon: 119.6487W

Elev: 1490

StdP: 13.92

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 6.9        | 12.4 | -3.7                          | 4.8 | 8.3  | 1.9 | 6.4 | 14.2 | 23.4                  | 40.9 | 19.8 | 40.3 | 7.2                      | 290  | 0.500 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.3                   | 91.3            | 63.4 | 88.1 | 62.6 | 84.8 | 61.6 | 66.4                | 84.5 | 64.6 | 82.3 | 63.0 | 80.5 | 5.8                  | 40   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.2                            | 82.1 | 71.9 | 58.0 | 75.9 | 70.5 | 55.9 | 70.4 | 69.3 | 31.8          | 84.6 | 30.4 | 82.1 | 29.2 | 80.6 | 72.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |  |  |
| (a)               | (b)  | (c)  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |  |  |
| 18.0              | 14.3 | 11.8 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |  |  |
|                   |      |      | 2.6                        | 97.4 | 5.0                | 2.5 | -1.0                                               | 99.1 | -3.9       | 100.6 | -6.7       | 102.0 | -10.3      | 103.8 |  |  |
|                   |      |      | 0.6                        | 68.9 | 4.4                | 2.4 | -2.5                                               | 70.6 | -5.1       | 72.0  | -7.5       | 73.4  | -10.7      | 75.1  |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.2   | 29.7  | 32.4  | 40.5  | 48.3  | 57.4  | 63.6  | 71.3  | 70.2  | 60.6  | 48.0  | 37.2  | 29.9 | (6)  |
|                                                                        | DBStd       | 16.19  | 8.18  | 7.52  | 6.82  | 5.76  | 6.66  | 6.16  | 6.16  | 5.82  | 5.96  | 5.82  | 7.76  | 7.75 | (7)  |
|                                                                        | HDD50       | 2653   | 630   | 494   | 299   | 99    | 12    | 0     | 0     | 2     | 106   | 387   | 624   |      | (8)  |
|                                                                        | HDD65       | 6264   | 1095  | 914   | 760   | 500   | 251   | 98    | 16    | 21    | 159   | 527   | 834   | 1089 | (9)  |
|                                                                        | CDD50       | 2353   | 0     | 0     | 4     | 49    | 241   | 407   | 661   | 626   | 319   | 44    | 2     | 0    | (10) |
|                                                                        | CDD65       | 489    | 0     | 0     | 0     | 0     | 14    | 55    | 212   | 182   | 26    | 0     | 0     | 0    | (11) |
|                                                                        | CDH74       | 5352   | 0     | 0     | 0     | 5     | 206   | 635   | 2400  | 1859  | 247   | 0     | 0     | 0    | (12) |
|                                                                        | CDH80       | 1988   | 0     | 0     | 0     | 0     | 44    | 194   | 1007  | 687   | 56    | 0     | 0     | 0    | (13) |
|                                                                        | WSAvg       | 5.3    | 4.8   | 5.1   | 5.6   | 5.4   | 5.1   | 5.1   | 5.2   | 5.3   | 5.3   | 5.6   | 5.9   | 5.2  | (14) |
|                                                                        | PrecAvg     | 12.6   | 1.2   | 0.8   | 0.7   | 1.0   | 1.3   | 1.3   | 1.1   | 1.2   | 0.9   | 0.7   | 1.1   | 1.3  | (15) |
| (16) Precipitation                                                     | PrecMax     | 19.4   | 2.6   | 2.0   | 1.6   | 2.4   | 3.7   | 4.4   | 2.8   | 3.2   | 2.1   | 2.5   | 2.6   | 2.8  | (16) |
|                                                                        | PrecMin     | 8.4    | 0.2   | 0.2   | 0.1   | 0.2   | 0.3   | 0.3   | 0.0   | 0.1   | 0.1   | 0.2   | 0.4   | 0.4  | (17) |
|                                                                        | PrecStd     | 2.6    | 0.6   | 0.5   | 0.4   | 0.7   | 0.7   | 0.9   | 0.7   | 0.7   | 0.6   | 0.5   | 0.6   | 0.7  | (18) |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |      |      |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 47.0   | 50.8  | 61.9  | 73.4  | 85.2  | 90.7  | 96.7  | 94.3  | 85.5  | 68.7  | 57.2  | 49.0  |      | (19) |
|                                                                        | 0.4% MCWB   | 40.6   | 43.2  | 47.7  | 54.8  | 60.3  | 62.9  | 65.7  | 63.9  | 61.8  | 55.3  | 49.9  | 43.6  |      | (20) |
|                                                                        | 2% DB       | 43.6   | 47.1  | 57.8  | 67.7  | 79.6  | 85.5  | 92.5  | 90.2  | 80.4  | 64.1  | 52.9  | 44.7  |      | (21) |
|                                                                        | 2% MCWB     | 39.2   | 40.6  | 45.7  | 51.2  | 57.7  | 60.8  | 63.6  | 63.0  | 60.1  | 52.3  | 46.5  | 39.7  |      | (22) |
|                                                                        | 5% DB       | 41.1   | 44.3  | 54.3  | 63.8  | 75.0  | 81.0  | 89.5  | 87.1  | 76.2  | 60.7  | 49.7  | 41.7  |      | (23) |
|                                                                        | 5% MCWB     | 37.7   | 39.0  | 44.0  | 48.9  | 55.9  | 59.5  | 63.1  | 62.7  | 58.7  | 50.3  | 44.6  | 37.9  |      | (24) |
|                                                                        | 10% DB      | 38.6   | 41.8  | 51.1  | 60.0  | 70.9  | 76.9  | 85.9  | 83.8  | 72.6  | 57.9  | 47.3  | 39.0  |      | (25) |
|                                                                        | 10% MCWB    | 36.1   | 37.4  | 42.1  | 46.9  | 53.9  | 58.1  | 62.2  | 61.7  | 56.8  | 48.7  | 42.7  | 36.0  |      | (26) |
|                                                                        | 0.4% WB     | 41.8   | 45.0  | 49.7  | 55.6  | 61.9  | 65.6  | 69.3  | 68.8  | 64.7  | 57.2  | 50.5  | 44.4  |      | (27) |
|                                                                        | 0.4% MCDB   | 45.1   | 49.3  | 58.4  | 70.3  | 78.8  | 83.3  | 89.5  | 87.8  | 79.1  | 65.5  | 55.3  | 48.0  |      | (28) |
| (30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 39.5   | 41.9  | 47.1  | 52.5  | 59.4  | 63.0  | 66.8  | 66.4  | 62.2  | 54.5  | 47.4  | 40.8  |      | (29) |
|                                                                        | 2% MCDB     | 42.5   | 45.4  | 55.5  | 64.9  | 74.7  | 80.0  | 85.7  | 83.9  | 76.0  | 61.5  | 51.3  | 43.9  |      | (30) |
|                                                                        | 5% WB       | 37.9   | 39.8  | 45.0  | 50.1  | 57.5  | 61.1  | 65.0  | 64.4  | 60.1  | 52.1  | 45.4  | 38.4  |      | (31) |
|                                                                        | 5% MCDB     | 40.4   | 43.4  | 52.4  | 61.2  | 72.1  | 76.9  | 83.5  | 81.6  | 73.5  | 59.0  | 48.9  | 41.0  |      | (32) |
|                                                                        | 10% WB      | 36.3   | 37.9  | 43.1  | 48.1  | 55.5  | 59.3  | 63.4  | 62.8  | 58.1  | 49.9  | 43.0  | 36.2  |      | (33) |
|                                                                        | 10% MCDB    | 38.3   | 41.1  | 50.1  | 58.1  | 68.3  | 74.0  | 81.6  | 80.1  | 70.5  | 56.6  | 46.4  | 38.7  |      | (34) |
| (36) Mean Daily Temperature Range                                      | MDBR        | 8.6    | 12.2  | 16.1  | 20.0  | 21.8  | 21.6  | 24.3  | 23.8  | 20.7  | 15.8  | 10.5  | 8.5   |      | (35) |
|                                                                        | 5% DB MCDBR | 10.6   | 13.9  | 20.2  | 25.4  | 28.1  | 27.6  | 29.5  | 28.3  | 25.5  | 18.9  | 12.9  | 10.6  |      | (36) |
|                                                                        | 5% DB MCWBR | 7.7    | 9.1   | 11.6  | 13.2  | 13.1  | 12.4  | 12.5  | 12.7  | 13.1  | 11.0  | 8.9   | 7.8   |      | (37) |
|                                                                        | 5% WB MCDBR | 9.8    | 12.7  | 18.3  | 22.7  | 24.5  | 24.5  | 26.2  | 25.6  | 23.4  | 16.9  | 11.9  | 9.9   |      | (38) |
|                                                                        | 5% WB MCWBR | 7.4    | 8.7   | 11.3  | 12.7  | 12.2  | 11.6  | 12.3  | 12.4  | 12.7  | 10.6  | 8.5   | 7.6   |      | (39) |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |      |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.291  | 0.290 | 0.318 | 0.343 | 0.374 | 0.366 | 0.356 | 0.366 | 0.340 | 0.330 | 0.305 | 0.280 |      | (40) |
|                                                                        | taud        | 2.330  | 2.388 | 2.408 | 2.403 | 2.350 | 2.400 | 2.454 | 2.429 | 2.486 | 2.490 | 2.458 | 2.398 |      | (41) |
|                                                                        | Ebn at Noon | 240    | 274   | 284   | 287   | 280   | 283   | 285   | 276   | 273   | 255   | 235   | 230   |      | (42) |
|                                                                        | Edh at Noon | 22     | 26    | 30    | 34    | 37    | 36    | 34    | 33    | 28    | 23    | 19    | 18    |      | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 321    | 619   | 982   | 1449  | 1722  | 1810  | 2089  | 1780  | 1259  | 719   | 338   | 244   |      | (44) |
|                                                                        | RadStd      | 37     | 63    | 95    | 110   | 136   | 163   | 167   | 130   | 110   | 89    | 41    | 28    |      | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | -1.04   | -1.42 | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (18 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page